

FAIR-PLACO

Towards Fairness-Aware Human-AI Team Formation

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Human-AI teams improve accuracy — but ignore fairness.

COMHAI, PLACO optimise human selection for **accuracy**, not fairness

Accuracy-optimal routing can:

- disadvantage minority groups
- amplify model bias
- hide subgroup unfairness

Results show:

- unbiased teams achieve 96–97% accuracy
- fairness does not improve automatically
- human bias is the main fairness bottleneck

Goal: fairness-aware team formation.

Algorithm 1 PLACO Greedy Subset Selection

Require: Instance $x \in \mathcal{X}$, human set $\mathcal{H}$, cost function C

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1: Let  $subset \leftarrow \{\}$ ,  $subset\_cost \leftarrow 0$ 
2: Let  $max\_human \leftarrow -1$ ,  $max\_value \leftarrow -1$ 
3: Let  $y^* \leftarrow \arg \max_{j \in \mathcal{Y}} \prod_{i \in \mathcal{H}} V_i(x, j)$ 
4: for  $i \leftarrow 1$  to  $|\mathcal{H}|$  do
5:    $human\_value \leftarrow V_i(x, y^*)$ 
6:   if  $human\_value > max\_value$  then
7:      $max\_value \leftarrow human\_value$ 
8:      $max\_human \leftarrow i$ 
9:   end if
10:  if  $human\_value > 1$  then
11:     $subset \leftarrow subset \cup i$ 
12:     $subset\_cost \leftarrow subset\_cost + C(i)$ 
13:  end if
14: end for
15: if  $subset$  is empty then
16:    $subset \leftarrow subset \cup max\_human$ 
17:    $subset\_cost \leftarrow subset\_cost + C(max\_human)$ 
18: end if
19: Output  $subset, subset\_cost$ 
```

PLACO GreedySubsetSelection algorithm, the core framework
extended in this work for fairness-awareness

Two complementary levers:

1. Fair AI Prior (FairAI)

Replace the base classifier with ExponentiatedGradient trained under an Equalized Odds constraint. Reduces the fairness gap *before* any human is consulted.

2. Lagrangian Routing

Modify PLACO's value function to penalise human subset choices that worsen running fairness violations:

$$\log V^{\text{fair}} = \underbrace{\log V^{\text{PLACO}}}_{\text{accuracy}} - \underbrace{K \sum_m \lambda_m \max(0, \Delta \Psi_m)}_{\text{fairness penalty}} - \underbrace{\beta C(S)}_{\text{cost}}$$

Dual weights λ_m are updated online every 64 instances.

Three human bias regimes:

Unbiased — random label noise, no group dependence

Explicitly biased — hard-deny or accuracy penalty on minority instances when AI is uncertain

Automation-biased — copies AI label 85% of the time ($\tau = 0.85$)

Fairness metrics tracked:

DP, EO, EOpp, TE, Subgroup Fairness, Individual Fairness

12-Setting Grid

	AI / Policy	Bias
G1	BasicAI + PLACO	U/E/A
G2	FairAI + PLACO	U/E/A
G3	FairAI + FAIR-PLACO	U/E/A
G4	BasicAI + FAIR-PLACO	U/E/A

U = Unbiased E = Explicit A = Automation

6 Datasets

COMPAS · Adult · German Credit
ACS Income · CIFAR-10H ·
ImageNet-16H

15 humans per run, budget = 30%,
costs $\sim \mathcal{U}(0, 2)$

G1 = baseline G2 = fair AI only
G3 = full system G4 = routing
ablation

FAIR-PLACO: Fairness-Aware Human-AI Team Formation

4x3 summary: Accuracy vs Equalized-Odds gap across (group × human-bias) combinations



Results — Average Metrics Across All Datasets

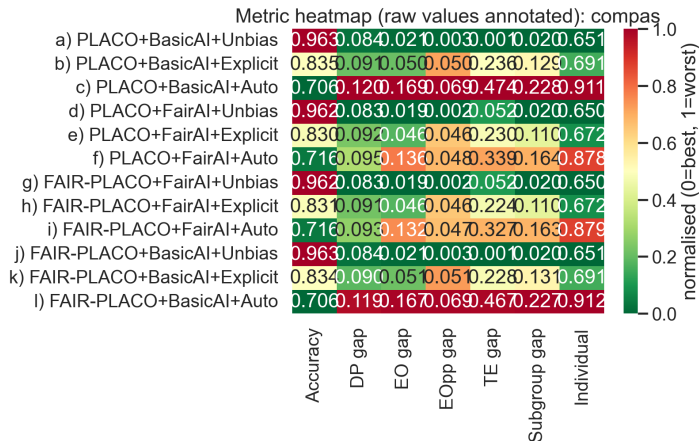
Setting	Acc↑	DP↓	EO↓	EOpp↓	TE↓	Subgrp↓	IndivF↑
a) PLACO + BasicAI + Unbiased	0.970	0.096	0.023	0.015	0.243	0.067	0.700
b) PLACO + BasicAI + Explicit	0.884	0.091	0.197	0.162	0.309	0.322	0.719
c) PLACO + BasicAI + Auto	0.769	0.107	0.156	0.122	0.382	0.436	0.847
d) PLACO + FairAI + Unbiased	0.969	0.095	0.019	0.012	0.228	0.069	0.698
e) PLACO + FairAI + Explicit	0.878	0.082	0.199	0.157	0.537	0.319	0.708
f) PLACO + FairAI + Auto	0.773	0.074	0.122	0.088	0.230	0.394	0.840
g) FAIR-PLACO + FairAI + Unbiased	0.967	0.093	0.022	0.013	0.299	0.079	0.698
h) FAIR-PLACO + FairAI + Explicit	0.878	0.081	0.199	0.158	0.554	0.323	0.706
i) FAIR-PLACO + FairAI + Auto	0.772	0.074	0.122	0.087	0.228	0.393	0.841
j) FAIR-PLACO + BasicAI + Unbiased	0.968	0.094	0.024	0.017	0.293	0.078	0.700
k) FAIR-PLACO + BasicAI + Explicit	0.883	0.089	0.198	0.163	0.332	0.324	0.719
l) FAIR-PLACO + BasicAI + Auto	0.769	0.107	0.155	0.121	0.379	0.435	0.847

d leads — best EO/EOpp in unbiased regime; fair AI prior is the single most effective lever.

g adds targeted EOpp wins on top of d; accuracy difference is negligible.

Bias rows spike sharply — human behaviour dominates fairness outcomes regardless of model.

Analysis — Metric Heatmaps



COMPAS: full 12-setting × 7-metric matrix

How to read it

Rows = 12 settings; columns = 7 tracked metrics. Values are normalised per metric.

What it reveals

d & g consistently minimise EO/EOpp while preserving peak accuracy

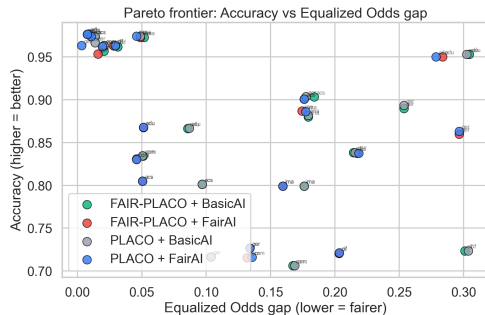
Explicit bias creates the sharpest fairness failures, concentrated in EO and Subgroup metrics

Automation bias propagates model errors, degrading both fairness and reliability

j shows routing helps, but cannot repair an unfair base classifier

Heatmaps expose metric-level failures hidden by averages.

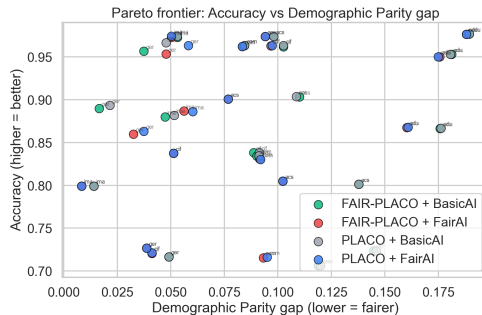
Results — Accuracy–Fairness Pareto Frontier



Accuracy vs. Equalized Odds gap

Settings d & g dominate the frontier — top-left (high accuracy, low fairness gap).

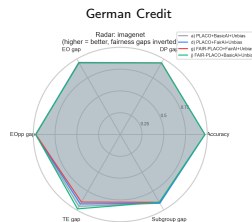
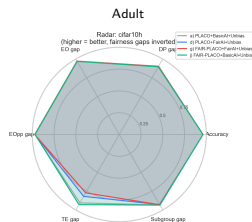
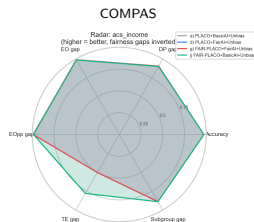
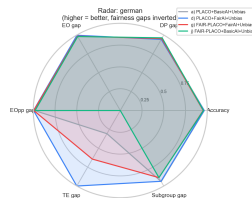
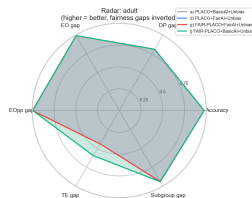
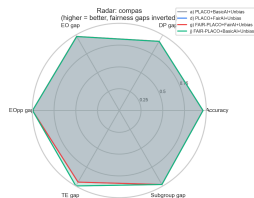
The **accuracy cost** of fairness constraints is **under 0.3%** — fairness and accuracy are not fundamentally in conflict.



Accuracy vs. Demographic Parity gap

Biased regimes scatter bottom-right — no policy recovers from systematically biased annotators.

Results — Per-Dataset Radar Charts (Unbiased Setting)



Key observations:

d & g are nearly identical — the fair prior drives most fairness gains.

j (routing only) weakens on Subgroup & TE — routing cannot fully offset an unfair AI prior.

German Credit shows the largest spread — strongest accuracy–fairness trade-off.

Image datasets are tightly clustered — synthetic sensitive attributes limit fairness sensitivity.

Core Takeaways

Fair AI prior is the single most effective lever — EO ↓14% at <0.1% accuracy cost

Lagrangian routing adds complementary EOpp gains; works even with a standard AI

Human bias dominates — no algorithm fully recovers from systematically biased annotators

Automation-biased humans *paired with a fair AI* can **amplify fairness** — a novel design insight

Open Directions

Online confusion matrix updates — human behaviour adapts; fixed matrices become stale

Structured cost models — real annotator cost correlates with expertise

Multi-class & intersectional fairness — binary *A* is a strong simplification

Bias-resilient human selection — filter or re-weight biased annotators at routing time

Adaptive dual batch size — German Credit shows instability at small scale

FAIR-PLACO is a **modular template** — any combination method, fairness metric, or subset-selection algorithm can be plugged in.